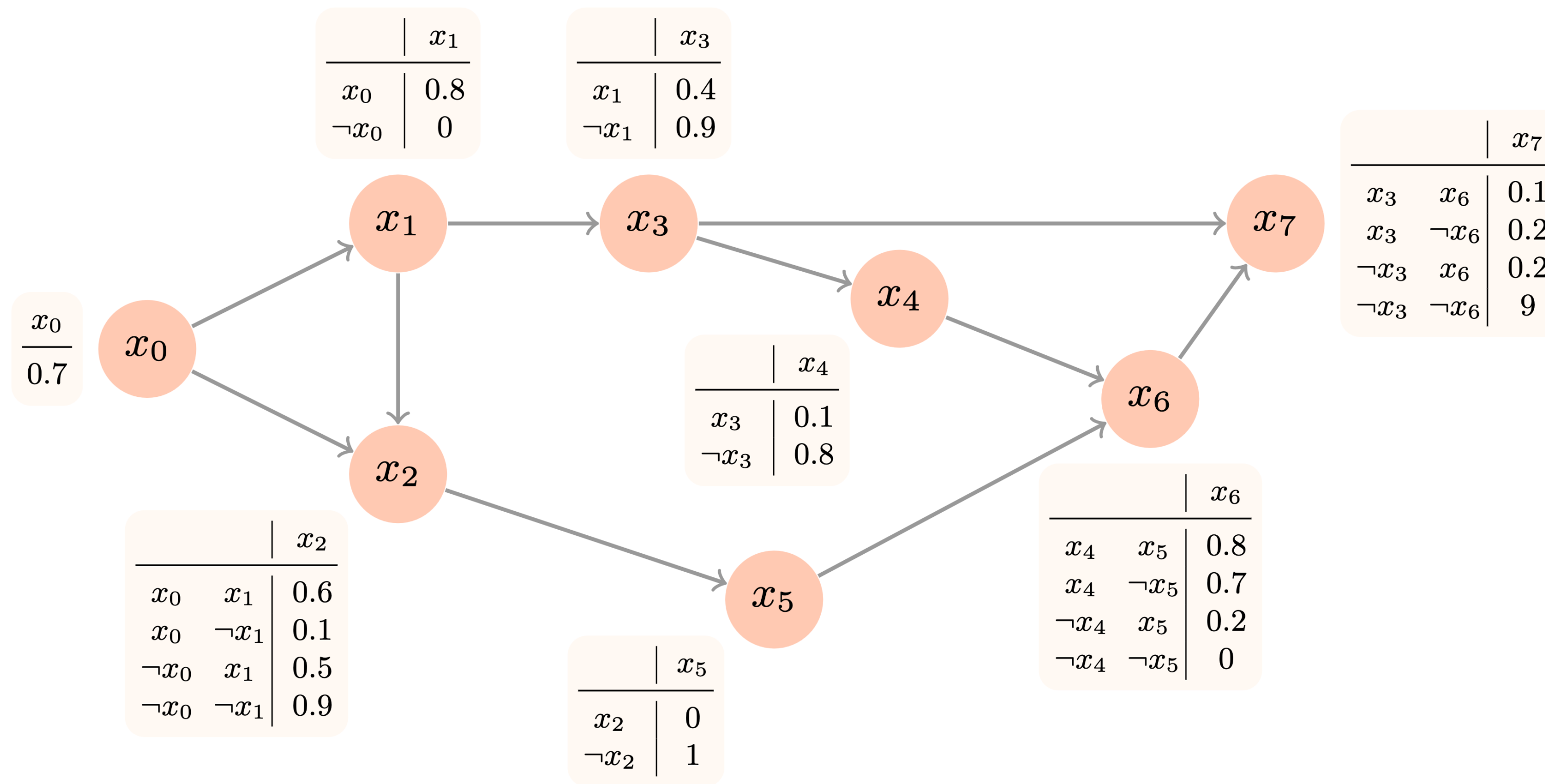


Probabilistic Ontologies

Bayesian Reasoning in Protégé

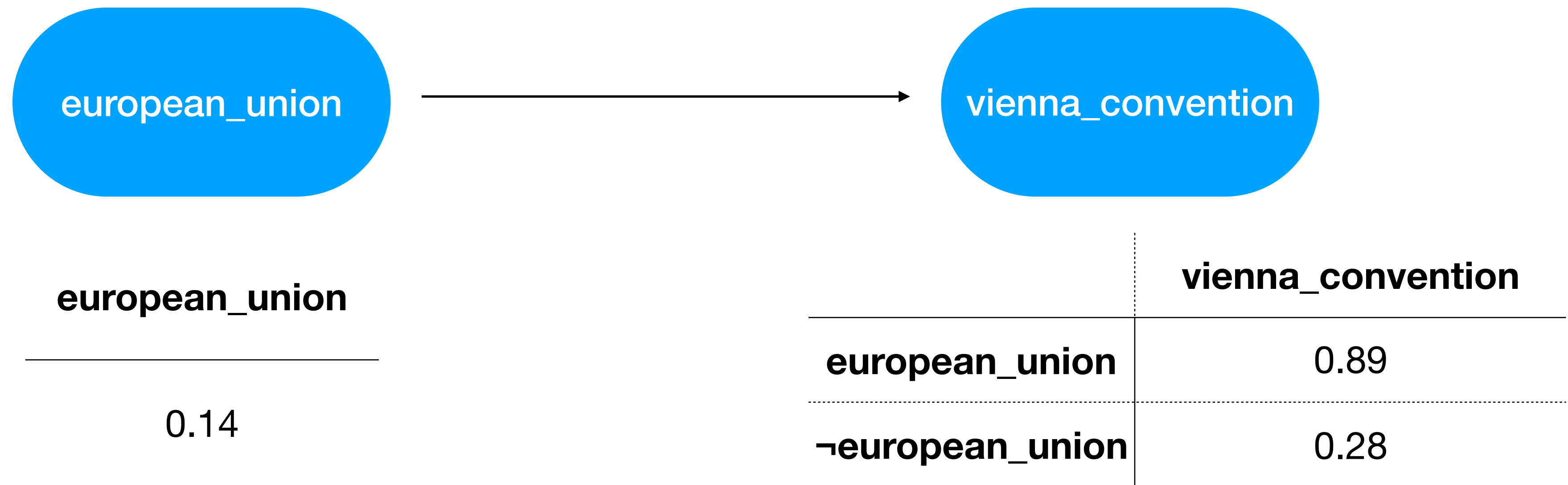
- BORN – a Bayesian ontology reasoner (<https://julianmendez.github.io/born/>)
- Protégé plugin has been uploaded to the Campus Virtual: needs to be manually uploaded into the ‘plugins’ folder of your Protégé installation.
- The BORN reasoner is executed in a dedicated tap, and Python 3 has to be executable from your path
- Two files:
 - $\mathcal{EL}$ ontology whose GCIs may be annotated with *context* variables
 - Bayesian network of contexts, written in Prolog-like syntax (ProbLog)

Bayesian Network



$0.7 :: x_0.$
 $0.8 :: x_1 :- x_0.$
 $0 :: x_1 :- \backslash + x_0.$
 $0.6 :: x_2 :- x_0, x_1.$
 $0.1 :: x_2 :- x_0, \backslash + x_1.$
 $0.5 :: x_2 :- \backslash + x_0, x_1.$
 $0.9 :: x_2 :- \backslash + x_0, \backslash + x_1.$

Road Sign Contexts



untitled-ontology-402 (http://www.semanticweb.org/schorlemmer/ontologies/2026/4/untitled-ontology-402) : [/Users/schorlemmer/Desktop/foo2.owl]

< >

untitled-ontology-402 (http://www.semanticweb.org/schorlemmer/ontologies/2026/4/untitled-ontology-402) : [/Users/schorlemmer/Desktop/foo2.owl]

◇ 🔍

> DiamondShapedSign > DangerWarningSign

Active ontology × BORN × Entities × Individuals by class × DL Query × SPARQL

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: DangerWarningSign ? 📄 📄 📄 📄

👤 + 📄 + ✖ +

Asserted ▾

owl:Thing

DiamondShapedSign

DangerWarningSign

TriangleShapedSign

DangerWarningSign

Annotations Usage

Annotations: DangerWarningSign

Annotations +

Description: DangerWarningSign

Equivalent To +

SubClass Of +

DiamondShapedSign

TriangleShapedSign

General class axioms +

SubClass Of (Anonymous Ancestor)

Instances +

Target for Key +

Disjoint With +

Disjoint Union Of +

Annotations for SubClassOf

DangerWarningSign SubClassOf TriangleShapedSign

Annotations +

prob

vienna_convention

Annotations +

OK

Annotations +

Annotations: DangerWarningSign

Annotations +

Description: DangerWarningSign

Equivalent To +

SubClass Of +

DiamondShapedSign

TriangleShapedSign

General class axioms +

SubClass Of (Anonymous Ancestor)

Instances +

Target for Key +

Disjoint With +

Disjoint Union Of +

To use the reasoner click Reasoner > Start reasoner

☒ Show Inferences

⚠

untitled-ontology-402 (http://www.semanticweb.org/schorlemmer/ontologies/2026/4/untitled-ontology-402) : [/Users/schorlemmer/Desktop/foo2.owl]

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untitled-ontology-402 (http://www.semanticweb.org/schorlemmer/ontologies/2026/4/untitled-ontology-402)

> prob

Active ontology x BORN x Entities x Individuals by class x DL Query x SPARQL Query x

Annotation properties

Datatypes

Individuals

Classes

Object properties

Data properties

Annotation property hierarchy: prob

owl:backwardCompatibleWith

owl:deprecated

owl:incompatibleWith

owl:priorVersion

owl:versionInfo

prob

rdfs:comment

rdfs:isDefinedBy

rdfs:label

rdfs:seeAlso

Annotations

Usage

Annotations: prob

Annotations +

Description: prob

Domains (intersection) +

Range (intersection) +

Superproperties +

To use the reasoner click Reasoner > Start reasoner

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untitled-ontology-402 (http://www.semanticweb.org/schorlemmer/ontologies/2026/4/untitled-ontology-402) : [/Users/schorlemmer/Desktop/foo2.owl]

untitled-ontology-402 (http://www.semanticweb.org/schorlemmer/ontologies/2026/4/untitled-ontology-402)

Active ontology x Entities x Individuals by class x DL Query x SPARQL Query x BORN x

BORN:

Reasoner Tests Experiments

ONTOLOGY

/Users/schorlemmer/Desktop/foo3.owl

BAYESIAN NETWORK

/Users/schorlemmer/Desktop/bn.pl

RULES

con(A1),
con(A2),
con(B).
sub(X, exists(R, B)) :-
 gci(A, exists(R, B)),
 sub(X, A),
 con(X),
 con(A),
 con(B),
 role(R).
sub(X, B) :-
 gci(exists(R, A), B),
 sub(X, exists(R, Y)),
 sub(Y, A),
 con(X),
 con(Y),
 con(A),
 con(B),
 role(R).

% Rules to avoid empty predicates of entities
con(-).
role(-).
indiv(-).

EXAMPLES

example-0

QUERY

query(sub(':DangerWarningSign',':DiamondShapedSign')).

RESULT

sub(':DangerWarningSign',':DiamondShapedSign'):0

Class hierarchy:

Asserted

Description:

Equivalent To

To use the reasoner click Reasoner > Start reasoner

Show Inferences

References

- I. I. Ceylan & R. Peñaloza
The Bayesian Description Logic *BEL*
IJCAR 2014, LNAI 8562, pp. 480-494, 2014
https://doi.org/10.1007/978-3-319-08587-6_37
- I.I. Ceylan, J. Mendez, & R. Peñaloza
The Bayesian Ontology Reasoner is BORN!
https://ceur-ws.org/Vol-1387/paper_5.pdf